ARAM: A Lightweight Adaptive Rectangular Attention Mamba for Hyperspectral Image Classification

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***Abstract*—Recently, Mamba has gradually been applied to hyperspectral image classification (HSIC). However, Mamba's ability to extract local detail features is relatively weak. Additionally, due to its causal nature, most existing methods perform multidirectional scanning, which further increases computational complexity. To overcome these challenges, this article presents a new lightweight adaptive rectangular attention Mamba (ARAM) network for HSIC. Our model fully leverages both local and global features of hyperspectral images (HSIs), and consists of an adaptive rectangular convolution (ARConv) module for extracting local features and an attention Mamba (AM) module for extracting global features. Specifically, we first designed ARConv, where, instead of using fixed-size convolution kernels, the width and height of the kernels are learned adaptively as parameters. This enhances the model's proficiency in capturing fine local details at different scales. The AM module is adopted to mitigate the inherent causal limitations and long-range decay characteristics of Mamba. An attention state-space equation (ASE) is also incorporated to model the state space in the semantic space, alleviating long-range decay issues with global perception capabilities. Furthermore, a new semantic-guided neighborhood (SGN) mechanism is constructed to reorganize features based on similar semantics, enhancing effective interaction between distant but similar features. Experimental results on multiple datasets demonstrate that ARAM achieves superior results compared with current state-of-the-art methods, substantially reducing the model size as well as the number of floating-point operations (FLOPs).**

***Index Terms*—Adaptive rectangular convolution (ARConv), hyperspectral image classification (HSIC), lightweight model, Mamba.**

I. Introduction

UNDER conventional observation conditions, the human visual system and traditional imaging devices are limited to the visible spectrum. Hyperspectral imaging technology breaks through this physical limitation, allowing spectral imaging instruments to continuously acquire multiband spectral information spanning the visible to long-wave infrared spectrum [1], [2], [3]. Hyperspectral images (HSIs) contain rich spatial–spectral information, enabling precise classification of land cover types at the pixel level [4], [5]. This unique advantage has garnered widespread attention for HSI classification (HSIC) techniques in practical applications [6], [7], [8]. At present, this technology has been widely deployed in various domains, including military reconnaissance [9], agricultural remote sensing [10], and mineral resource exploration [11], currently showing great potential for future development.

Initially, HSIC methods relied largely on machine learning and statistical principles, including support vector machines (SVMs) [12], principal component analysis (PCA) [13], linear discriminant analysis (LDA) [14], Random Forests [15], etc. However, these methods predominantly utilized the spectral features of HSIs while neglecting the rich spatial information contained in the data. To address this limitation, researchers introduced techniques such as superpixel segmentation [16], extended morphological profiles [17], and sparse manifold representation [18]. Nevertheless, these methods rely on predefined parameters, which limit their representational capacity in complex scenarios.

Benefiting from the impressive automatic feature extraction power offered by deep learning, widespread development has been seen in HSIC [19], [20]. Among these, convolutional neural networks (CNNs) have been extensively utilized, owing to their outstanding capacity for local feature extraction. For instance, Lee and Kwon [21] proposed a 2-D CNN that effectively captures the spatial correlations within a single HSI band. Ben Hamida et al. [22] developed a 3-D CNN that leverages 3-D convolution kernels to simultaneously extract both

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spatial and spectral local features. Yu et al. [23] introduced a spatial–spectral dense convolutional neural network (FAD-CNN), which enhances feature expression using high-level semantic data to guide the extraction of low-level features. Zhu et al. [24] proposed a lightweight feature extraction network built on efficient asymmetric dilated convolution (EADC) modules to capture multiscale spatial local information. While CNN-based methods excel at local feature extraction, the limited receptive field of local features presents challenges for the model's global modeling capability.

With the development of attention mechanisms, Transformers have broken through the limitations of existing CNN methods, which are confined to local receptive fields, enabling effective modeling of global contextual features in HSIs [25], [26]. Hong et al. [27] introduced SpectralFormer, a model dedicated to spectral sequence modeling to learn the sequential relationships between adjacent bands, thereby improving classification performance. To model the global spatial contextual dependencies in HSIs, Roy et al. [28] developed a morphological Spectral Transformer that combines morphological convolution and multihead self-attention to enhance the global spatial–spectral representation of HSI. In response to the inability of current Transformer networks to capture multiscale global spatial–spectral features, He et al. [29] introduced a multiscale spatial–spectral framework (HMSSF), a hybrid framework that collaboratively models global dependencies at different scales by designing spatial and spectral token generation modules and a multiscale self-attention mechanism. Liang et al. [30] further developed a Transformer that integrates the Fourier-wavelet transform and cross-stage edge enhancement modules to enhance global context interpretation and local edge information extraction from the frequency domain. Although the aforementioned Transformer-based methods exhibit significant advantages in terms of global modeling, the self-attention mechanism suffers from quadratic growth in computational complexity as the number of tokens increases, which challenges computational efficiency.

In recent years, Mamba, a novel deep learning framework built on selective structured state space models (S6), has gradually been applied to HSIC. This model achieves linear computational complexity for sequence modeling while significantly enhancing the ability to model long-range dependencies [31], [32]. Li et al. [33] pioneered the integration of Mamba into HSIC, proposing the new MambaHSI framework, which effectively captures long-range dependencies across HSIs while retaining linear complexity. Wang et al. [34] developed S^2Mamba, which utilizes spatial patch four-way scanning and spectral bidirectional scanning to efficiently model spatial–spectral contextual dependencies. To elevate the local feature extraction capability of Mamba, Liang et al. [35] introduced a Mamba-Like linear attention module, combined with depthwise separable convolutions for global–local modeling. Du et al. [36] proposed the Mamba-CNN knowledge distillation network (MKDnet), which improves cross-domain HSIC performance by simultaneously capturing both global and local features. He et al. [37] introduced HSI-Mformer, which merges the short-range modeling capabilities of Transformer with the long-range linear complexity modeling of Mamba to explore the complementarity between spatial–spectral global and local features. However, due to Mamba's reliance on causal state space modeling, it is prone to the long-range decay problem, which limits its global modeling capability.

In light of the above-mentioned issues, this article presents a novel lightweight adaptive rectangular attention Mamba model (ARAM) for HSIC, which simultaneously models global and local features with linear complexity. The framework mainly consists of an adaptive rectangular convolution (ARConv) module and an attention Mamba (AM) module. The ARConv module is introduced to address the challenge of fixed receptive fields in traditional convolutions by considering the height and width of convolution kernels as trainable parameters, enabling dynamic adjustment of the kernel's shape and orientation according to object size and sampling points. It allows for adaptive convolution and significantly enhances the representation of detailed pieces of information such as edges, linear land covers, and texture features. To strengthen Mamba's ability to model global context, we propose the AM module, which incorporates the attention state space equation (ASE) and the semantic-guided neighborhood (SGN) mechanism as key components. Specifically, ASE leverages Mamba's prompt learning to query semantically similar features, merging similar pixel sets across the entire image, enhancing semantic awareness, and alleviating long-range decay issues. SGN assigns semantic labels to each pixel, reconstructing the image to generate semantically adjacent 1-D sequences for scanning, facilitating effective interaction between distant but similar features. Compared with existing methods, the ARAM model strikes an optimal balance between classification performance and computational efficiency.

The core contributions of this work are summarized as follows.

1) We develop a novel ARAM framework for HSIC that enhances global and local feature extraction from HSIs with linear complexity.
2) An ARConv module is constructed, which dynamically tunes the shape and orientation of the convolution kernel according to the actual morphological features of land cover, improving the ability to capture local detail information in HSIs.
3) An ASE is designed, which queries information from semantically similar pixels through a prompt learning mechanism, alleviating the long-range decay issue.
4) A new SGN mechanism is proposed, which reorganizes features based on the semantic similarity between pixels, enhancing effective interaction between distant but similar features.

The remainder of this article is structured as follows. Section II presents a brief overview of related works. Section III provides a comprehensive description of the proposed framework. Section IV presents the experimental results and conducts corresponding analyses. Conclusions and future research avenues are discussed in Section V.

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II. Related Works

This section first reviews convolution operations, covering adaptive convolution and multiscale convolution, and introduces the proposed adaptive rectangular convolution. The fundamental principles of the Mamba selective scan model are then described, establishing the theoretical foundation for the proposed network detailed in Section III-B.

A. Convolution: Overview

1) Adaptive Convolution: Conventional standard convolution relies on fixed geometric shapes and receptive field sizes, which lack flexibility when dealing with complex geometric transformations, making it difficult to model multiscale and irregular objects commonly found in visual tasks. To alleviate this bottleneck, Dai et al. [38] developed deformable convolution, which first incorporates learnable offsets at standard spatial sampling locations. It empowers the model to adjust sampling positions adaptively according to task requirements and enables unsupervised deformation of the convolution kernel's geometric shape. Zhang et al. [39] proposed scale-adaptive convolution, which dynamically adjusts the receptive field by designing a scale regression layer, significantly improving the ability to recognize objects of various scales. Furthermore, researchers have developed methods such as dynamic snake convolution [40] and pixel-adaptive convolution (PAC) [41], further enhancing the feature extraction capabilities of adaptive convolutions.

Despite their effectiveness, these methods still suffer from notable limitations. On the one hand, highly flexible deformation mechanisms often require a large number of sampling points, resulting in increased computational overhead. On the other hand, constrained geometric deformation forms may fail to adequately capture complex or highly irregular object structures. Moreover, in most existing adaptive convolution methods, the number of sampling points is predefined and remains fixed during training, preventing dynamic adjustment alongside the learned kernel geometry. This constraint limits their adaptability to multiscale scenarios.

2) Multiscale Convolution: This kind of convolution enhances the perception of information at different scales by introducing convolution kernels of varying sizes to model the input features in parallel. For instance, [42] proposed the multiscale convolutional pooling network, which processes the input image with parallel convolutions of different kernel sizes, improving robustness to object scale variations. Cosmin Duta et al. [43] designed pyramid convolution, which introduces a pyramid structure of convolutional kernels with different sizes and depths in parallel within the same convolutional layer. This model overcomes the information loss and insufficient receptive field issues caused by standard convolutions that rely on small kernel stacking and downsampling to indirectly expand the receptive field while retaining high flexibility and scalability. To further expand the receptive field and keep a balance between extracting local details and global contextual cues, Yin et al. [44] combined dilated convolutions with different dilation rates and multiscale convolutions to construct the convolutional attention U-Net network.

Although the aforementioned methods enhance multiscale feature extraction capabilities, they predominantly rely on fixed combinations of convolutional kernels with predefined geometries. As a consequence, they lack the ability to adjust sampling locations and sampling point distributions dynamically, according to the geometric structures of different objects in the feature maps. This limitation restricts their capability to precisely model targets with significant variations in scale and shape.

3) Adaptive Rectangular Convolution: For HSIs characterized by complex structures and significant scale variations, standard convolutions with fixed receptive fields often struggle to adequately capture rich spatial–spectral representations. Although traditional adaptive shape convolution methods are capable of tuning sampling positions and the geometric structures of the kernels, the number of sampling points is usually predefined, limiting their ability to dynamically adapt to varying feature complexities. While multiscale convolution methods can extract multiscale features, they face challenges in dynamically adjusting the sampling positions.

In response to these constraints, the ARConv module is proposed in this article, which takes the height and width of convolution kernels as trainable parameters, thus enabling dynamic adjustment of kernel size and sampling point count according to land cover scale. Such a module allows ARAM to flexibly cope with the intrinsic complexity and multiscale spatial structure of HSI data, thereby enhancing its ability to extract local detail features.

B. State Space Models

State space models (SSMs) [45] are fundamental models in control theory for sequence modeling of continuous linear time-invariant (LTI) systems, and are typically mathematically formulated using the linear ordinary differential equation (ODE) below

$$\begin{cases} \boldsymbol{h}'(t) = \mathbf{A}\boldsymbol{h}(t) + \mathbf{B}\boldsymbol{x}(t) \\ \boldsymbol{y}(t) = \mathbf{C}\boldsymbol{h}(t) \end{cases} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{M\times M}$, $\mathbf{B}, \boldsymbol{h}'(t), \boldsymbol{h}(t) \in \mathbb{R}^{M\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times M}$. $\mathbf{A}$ denotes the state transition matrix, and $\mathbf{B}$ and $\mathbf{C}$ correspond to the input and output projection matrices, respectively. This expression illustrates the relationship between the 1-D function $x(t)$ and the output $y(t)$ mediated by the hidden state $\boldsymbol{h}(t)$.

In view of the fact that HSI data are generally discrete, the structured state space models (S4) were adopted [46], which transform SSMs, originally designed for continuous signals, into a formulation suitable for discrete sequential data through a discretization process. To discretize the continuous-time parameters $\mathbf{A}$ and $\mathbf{B}$, the zero-order hold (ZOH) method and the time-scale parameter Δ are applied, yielding the discrete counterparts $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. The discretization process is outlined as follows:

$$\begin{cases} \overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B} \end{cases} \tag{2}$$

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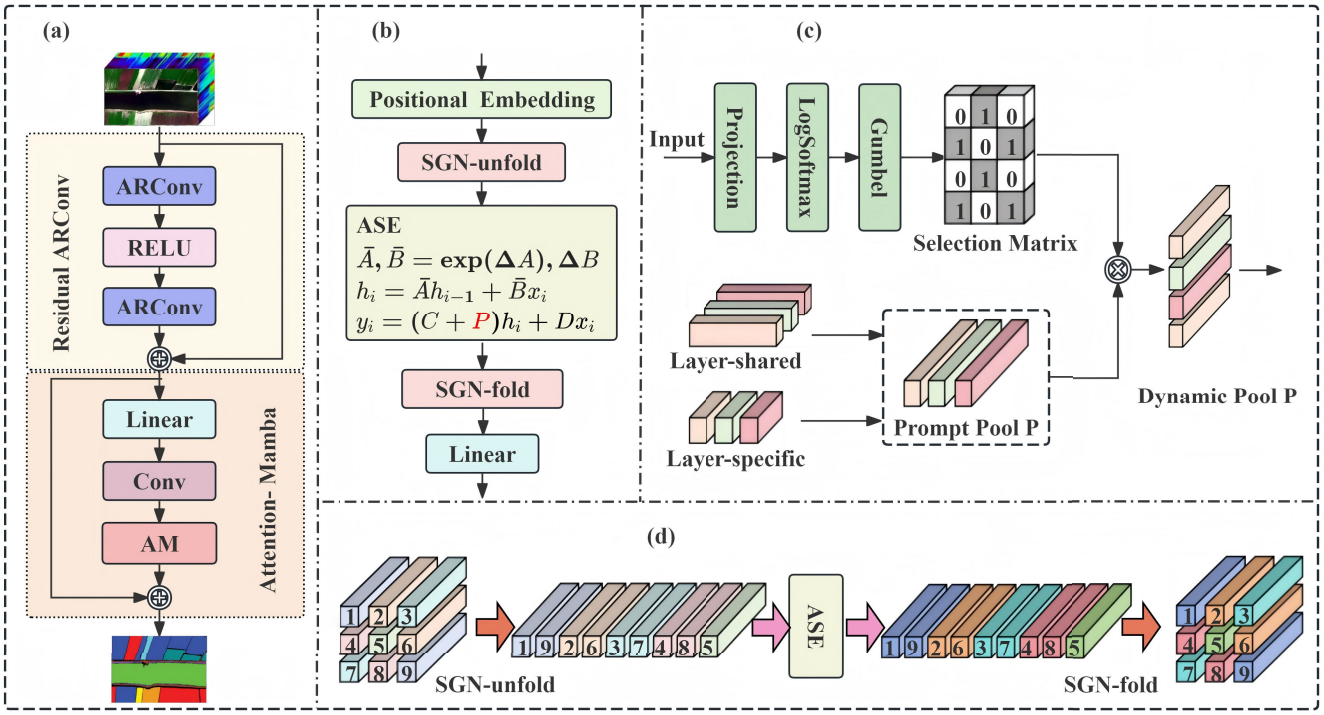


Fig. 1. (a) Overview of the ARAM framework, (b) AM, (c) ASE, and (d) SGN.

where $\mathbf{I}$ denotes the identity matrix. After discretization, the equation system is converted into the following formula:

$$\begin{cases} \boldsymbol{h}_t = \overline{\mathbf{A}}\boldsymbol{h}_{t-1} + \overline{\mathbf{B}}\boldsymbol{x}_t \\ \boldsymbol{y}_t = \mathbf{C}\boldsymbol{h}_t. \end{cases} \tag{3}$$

The process described above is a linear iterative process. In the S4 model, convolutional operations are employed to enhance its computational efficiency

$$\begin{cases} \overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{l-1}\overline{\mathbf{B}}\right) \\ \boldsymbol{y} = \boldsymbol{x} * \overline{\mathbf{K}} \end{cases} \tag{4}$$

where l stands for the length of the input sequence $\boldsymbol{x}$, with $\overline{\mathbf{K}} \in \mathbb{R}^T$ acting as the S4 convolution kernel, and * symbolizes the convolution operation.

The S4 model demonstrates strong capability in modeling long-range dependencies for deep learning sequence tasks while preserving linear computational complexity. Nevertheless, its parameters are globally fixed across the entire sequence, which limits its ability to adaptively emphasize informative elements and suppress irrelevant ones. To overcome this limitation, the work in [31] introduces a selective scan mechanism and proposes the selective scan-based S4 model, termed Mamba. Specifically, an input-dependent parameter adaptation mechanism is presented: the fixed parameter matrices $\mathbf{B}$, $\mathbf{C}$, and Δ of S4 are treated as dependent on the input data $\boldsymbol{x}$. By dynamically modulating these parameters at each sequence position, the model is able to selectively control the flow of information within the hidden state, effectively retaining task-relevant features while attenuating less informative components. This mechanism endows Mamba with content-adaptive sequence modeling capability analogous to attention, while preserving the efficiency advantages of state space models.

III. Proposed Method

In this section, we provide a detailed explanation of the proposed ARAM model, as depicted in Fig. 1. The primary components of this model include an ARConv residual module and an AM module. The ARConv module first processes the HSI data to extract local detailed features. Subsequently, the AM module enhances global pixel interactions for long-range dependency modeling, with the final classification results being output by the classifier.

A. Adaptive Rectangular Convolution

The core principle of ARConv is to dynamically adjust the height, width, and sampling points of the convolutional kernel, combined with bilinear interpolation and affine transformations, to adapt to multiscale objects. The process is divided into four steps, as illustrated in Fig. 2.

1) Learning the Height and Width Feature Maps of the Convolutional Kernel: Given $\mathbf{X} \in \mathbb{R}^{H\times W\times C_{\text{in}}}$, with H as height, W as width, and C_{in} as the number of input channels, the procedure for determining the height and width of the convolution kernel is outlined as

$$\boldsymbol{y}_i = \boldsymbol{f}_{\boldsymbol{\theta}_i}(\mathbf{X}), \quad \boldsymbol{i} \in \{1, 2\} \tag{5}$$

where the height h and width w are learned via two subnetworks $\boldsymbol{f}_{\boldsymbol{\theta}_i}$. As illustrated in Part (I) of Fig. 2, $\boldsymbol{f}_{\boldsymbol{\theta}_i}$ is realized by a dual-branch architecture that independently processes spatial

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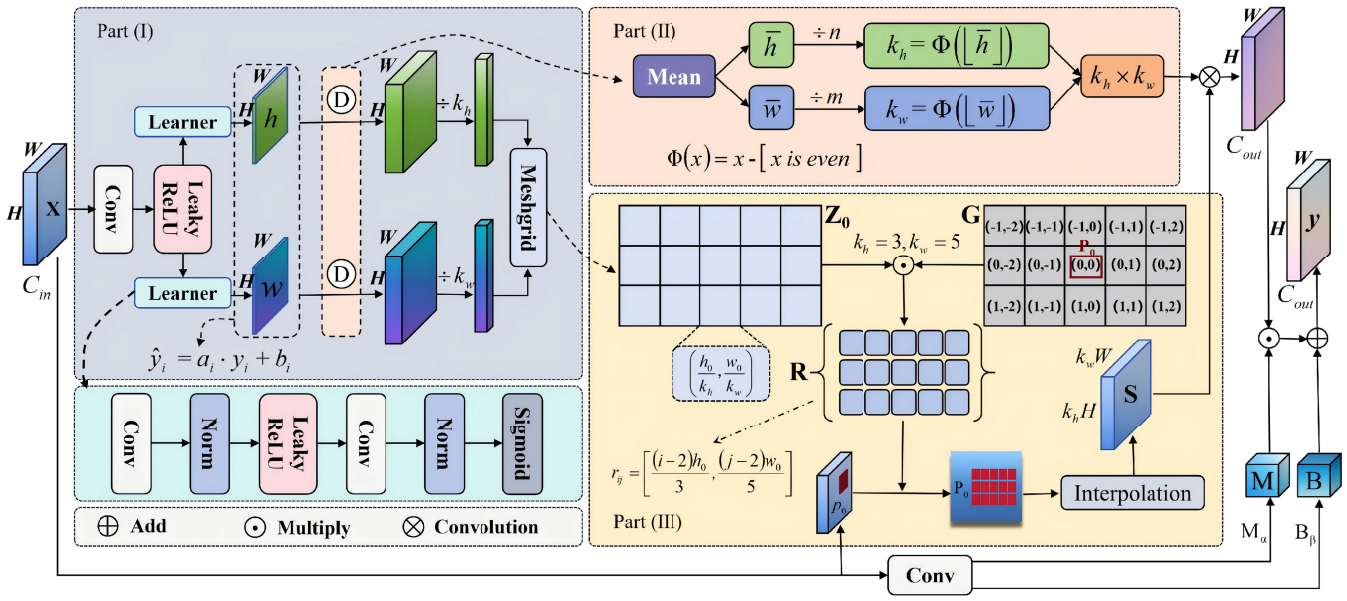


Fig. 2. Overview of the ARConv Module. This module is divided into four key components. Part (I) covers the learning of the convolution kernel's height and width feature maps. Part (II) addresses the process of selecting the number of sampling points for the convolution kernel. Part (III) illustrates the generation of the sampling map **S**, taking the grid center position $\mathbf{P}_0$ as an example. Part (IV) performs integrated processing of the trained features, as illustrated on the rightmost and lower-right sides. It consists of an affine transformation, representing convolution, element-wise multiplication, and element-wise addition.

features along the height and width axes. Each branch consists of a convolutional layer (Conv), a LeakyReLU activation function, and a Learner unit for adaptive feature learning. The output feature map $\boldsymbol{y}_i \in \mathbb{R}^{H\times W\times 1}$ consists of $\boldsymbol{y}_1$ for the height feature map and $\boldsymbol{y}_2$ for the width feature map, which correspond to $\boldsymbol{h}$ and $\boldsymbol{w}$ in Fig. 2, respectively. Specifically, features are initially captured through convolution. The extracted information is then fed into the dedicated Learner of each subnetwork, which performs learning and prediction based on the input features. Subsequently, the output convolution kernel's height and width are normalized using the Sigmoid function, yielding the relative size of the kernel, with y_i ranging from $(0, 1)$. Since y_i cannot directly correspond to the height and width, the range of the convolution kernel's height and width is constrained as follows:

$$\hat{y}_i = a_i \cdot y_i + b_i, \quad i \in \{1, 2\} \tag{6}$$

where a_i and b_i represent predefined scaling factors applied to limit the height and width ranges, respectively. Through this equation, h is restricted to the interval $(b_1, a_1 + b_1)$, while w is constrained to $(b_2, a_2 + b_2)$.

2) Selecting the Number of Sampling Points: The sampling points are determined by the convolution kernel size. Specifically, the vertical and horizontal sampling points are given by $k_h = \Phi(\lfloor \bar{h} \rfloor)$ and $k_w = \Phi(\lfloor \bar{w} \rfloor)$, where $\bar{h}$ and $\bar{w}$ are the average values of y_1 and y_2 obtained in the first step, and $\lfloor x \rfloor$ indicates the floor operation on x. m and n are modulation coefficients, and smaller values of these coefficients lead to a larger number of sampling points for a fixed kernel size. The function $\Phi(x)$ defines the mapping from the average height and width to the number of sampling points

$$\Phi(x) = x - [x \text{ is even}] \tag{7}$$

where $[\cdot]$ refers to the Iverson bracket: it evaluates to 1 if x is even, and to 0 if x is odd. In other words, if x is even, the function reduces it by 1, thereby ensuring that the number of sampling points is always odd. Through these operations, a total of $k_h \cdot k_w$ sampling points are finally obtained.

3) Generating the Sampling Map: The first two steps allow for dynamic learning of the convolution kernel size and the number of sampling points, and this step achieves adaptive adjustment of sampling positions by manipulating the sampling grid. In standard convolution, as the convolution kernel slides over the input feature map, it computes the weighted sum of the input region (i.e., the grid) at a specific position to generate the output pixel value at that position. This process is formalized as follows:

$$\boldsymbol{y}(\mathbf{P}_0) = \sum_{\boldsymbol{g}_n \in \mathbf{G}} \boldsymbol{w}(\boldsymbol{g}_n) \cdot \mathbf{X}(\mathbf{P}_0 + \boldsymbol{g}_n) \tag{8}$$

where $\boldsymbol{y}(\mathbf{P}_0)$ denotes the pixel value at position $\mathbf{P}_0$. $\mathbf{G}$ represents the grid containing all sampling points corresponding to the convolution kernel, g_n signifies the offset of a sampling point in $\mathbf{G}$ relative to position $\mathbf{P}_0$, $\mathbf{X}(\mathbf{P}_0 + \boldsymbol{g}_n)$ is the value at the position in the input $\mathbf{X}$ corresponding to the sampling point, and $\boldsymbol{w}(\cdot)$ denotes the parameters of the convolution kernel.

Notably, the convolution kernel in standard convolution samples the input feature map via a regular grid, with no gaps between adjacent sampling points. In contrast, ARConv generates a new sampling pattern by offsetting the center of the grid, which is implemented through the following steps.

1) *Defining the Standard Grid* $\mathbf{G}$*:* The offset matrix of standard convolution is represented by $\mathbf{G} \in \mathbb{R}^{k_h \times k_w}$, which stores the positions of all sampling points relative to the center of the convolution kernel. In particular, the element $\boldsymbol{g}_{ij}$ located at the ith row and jth column of $\mathbf{G}$

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is expressed as

$$\boldsymbol{g}_{ij} = \left(\frac{2i - k_h - 1}{2}, \frac{2j - k_w - 1}{2} \right). \tag{9}$$

2) *Generating the Scale Matrix* $\mathbf{Z_0}$*:* For each position $\mathbf{P}_0$, a scale matrix $\mathbf{Z_0} \in \mathbb{R}^{k_h \times k_w}$ is generated, which encodes the size of the convolution kernel at that position. The element z_{ij} located at the ith row and jth column of $\mathbf{Z_0}$ takes the form

$$\boldsymbol{z}_{ij} = \left(\frac{h_0}{k_h}, \frac{w_0}{k_w} \right) \tag{10}$$

where h_0 and w_0 denote the dimensions(height and width) of the convolution kernel learned at position $\mathbf{P}_0$.

3) *Compute the Offset Matrix* $\mathbf{R}$*:* Based on the standard grid $\mathbf{G}$ and the scale matrix $\mathbf{Z}_0$, the offset matrix $\mathbf{R}$ is obtained via an element-wise multiplication operation $\odot$

$$\mathbf{R} = \mathbf{Z}_0 \odot \mathbf{G} \tag{11}$$

where the matrix $\mathbf{R}$ stores the adaptively adjusted sampling positions (derived from the standard convolution grid), which are dynamically computed according to the input image features and the learned convolution kernel size. The value $\boldsymbol{r}_{ij}$ at the ith row and jth column of $\mathbf{R}$ can be written as

$$\boldsymbol{r}_{ij} = \left(\frac{(2i - k_h - 1)\, h_0}{2k_h}, \frac{(2j - k_w - 1)\, w_0}{2k_w} \right). \tag{12}$$

4) *Bilinear Interpolation:* Since sampling positions typically do not align with the grid centers, ARConv employs bilinear interpolation [47] to compute pixel values at the offset positions. The procedure is outlined as follows:

$$\boldsymbol{p}(x, y) = \boldsymbol{w}_x^{\top} \mathbf{P} \boldsymbol{w}_y \tag{13}$$

where $\boldsymbol{p}(x, y)$ indicates the pixel value at the point (x, y), $\boldsymbol{w}_x = (x - x_0)/(x_1 - x_0)$ and $\boldsymbol{w}_y = (y - y_0)/(y_1 - y_0)$ are the normalized interpolation weights along the x- and y-axes, respectively, and matrix $\mathbf{P}$ stores the pixel values of the four nearest grid points to (x, y), which are defined as

$$\mathbf{P} = \begin{pmatrix} \boldsymbol{p}(x_0, y_0) & \boldsymbol{p}(x_0, y_1) \\ \boldsymbol{p}(x_1, y_0) & \boldsymbol{p}(x_1, y_1) \end{pmatrix}. \tag{14}$$

5) *Generating the Sampling Map* $\mathbf{S}$*:* ARConv extracts the values corresponding to the sampling positions of each pixel's sampling window, and organizes them into a new grid $\mathbf{P}_0 \in \mathbb{R}^{k_h \times k_w \times C_{\text{in}}}$, which replaces the original pixel $\boldsymbol{p}_0$. By merging the grids $\mathbf{P}_0$ at all pixel positions, the final sampling map $\mathbf{S} \in \mathbb{R}^{(k_h H) \times (k_w W) \times C_{\text{in}}}$ is obtained. This sampling map contains the sampling points of all pixels in the original image, and it has been dynamically adjusted according to the image features.

4) Performing the Convolution: Subsequently, the convolution kernel is applied to the sampling map $\mathbf{S}$, where the kernel size and sliding step are defined as (k_h, k_w). The output feature $\boldsymbol{y}(\mathbf{P}_0)$ is computed via the following convolution process:

$$\boldsymbol{y}(\mathbf{P}_0) = \sum_{\boldsymbol{r}_n \in \mathbf{R}} \boldsymbol{w}(\boldsymbol{r}_n) \cdot \boldsymbol{p}(\mathbf{P}_0 + \boldsymbol{r}_n). \tag{15}$$

In contrast to (8), this operation updates the original offset g_n to the element r_n in the offset matrix $\mathbf{R}$, and replaces the original pixel value $\mathbf{X}(\mathbf{P}_0 + \boldsymbol{g}_n)$ with the new pixel value $\boldsymbol{p}(\mathbf{P}_0 + \boldsymbol{r}_n)$ (obtained after offset and interpolation operations), thus realizing the automatic adjustment of the convolution kernel's sampling points. To enable the feature map to adapt more flexibly to the spatial characteristics of the input image, two subnetworks $\mathbf{M}_\alpha$ and $\mathbf{B}_\beta$ are employed to predict the affine matrix $\mathbf{M}$ and the bias matrix $\mathbf{B}$, respectively: $\mathbf{M}_\alpha$ adjusts the spatial position of the convolution by learning the spatial structural features of the image; $\mathbf{B}_\beta$ performs a translation adjustment on the convolution result. Here, α and β stand for the subnetworks' parameters. The final output feature of the entire ARConv module is then given by

$$\boldsymbol{y} = \mathbf{SK} \otimes \mathbf{S} \odot \mathbf{M} \oplus \mathbf{B} \tag{16}$$

where $\boldsymbol{y} \in \mathbb{R}^{k_h \times k_w \times C_{\text{out}}}$, and $\mathbf{SK} \in \mathbb{R}^{C_{\text{in}} \times k_h \times k_w \times C_{\text{out}}}$ denotes the parameters of the convolution kernel, $\otimes$, $\odot$, and $\oplus$ represent convolution, element-wise multiplication, and element-wise addition, respectively.

B. Attention Mamba Module

Based on the mathematically established connection between SSM and Attention mechanisms, this article proposes an ASE that injects global semantic information into the output matrix $\mathbf{C}$ via residual addition, thereby enabling global noncausal modeling within the state-space framework. In addition, a new SGN mechanism is introduced to reconstruct image sequences according to semantic similarity, effectively establishing connections between spatially distant but semantically related pixels. This section first presents the mathematical relationship between SSM and Attention mechanisms, and then elaborates on the principles of the ASE and SGN mechanisms, as illustrated in Fig. 1.

1) Bridging State Space Models and Attention: The connection between SSM and Attention mechanisms provides a critical entry point for introducing noncausal modeling capability into Mamba [48]. In this section, the mathematical formulations of both paradigms are presented to investigate their underlying mathematical relationship.

We first analyze the Mamba model. For the discretized state-space equation given in (3), we perform the approximation $\overline{\mathbf{B}}\mathbf{x}_t \approx \Delta\mathbf{B}\mathbf{x}_t = \mathbf{B}(\Delta\mathbf{x}_t)$, which transforms (3) into

$$\begin{cases} \boldsymbol{h}_t = \overline{\mathbf{A}}\boldsymbol{h}_{t-1} + \mathbf{B}(\Delta\boldsymbol{x}_t) \\ \boldsymbol{y}_t = \mathbf{C}\boldsymbol{h}_t / \mathbf{I} + \mathbf{D}\boldsymbol{x}_t. \end{cases} \tag{17}$$

For the attention model, given the linear complexity and causal nature of Mamba, we draw an analogy with causal linear attention, and present the output form of causal linear attention as follows:

$$\boldsymbol{y}_t = \sum_{i=1}^{t} \frac{\mathbf{Q}_t \mathbf{K}_i^{\top}}{\sum_{j=1}^{t} \mathbf{Q}_t \mathbf{K}_j^{\top}} \mathbf{V}_i = \frac{\mathbf{Q}_t \left(\sum_{i=1}^{t} \mathbf{K}_i^{\top} \mathbf{V}_i \right)}{\mathbf{Q}_t \left(\sum_{j=1}^{t} \mathbf{K}_j^{\top} \right)} \tag{18}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote the query, key, and value matrices, which are key parameters in the attention mechanism.

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Subsequently, the matrix S_t is defined as

$$\mathbf{S}_t = \sum_{i=1}^{t} \mathbf{K}_i^\top \mathbf{V}_i, \quad \mathbf{Z}_t = \sum_{j=1}^{t} \mathbf{K}_j^\top. \tag{19}$$

Then, (18) can then be rewritten as

$$\begin{cases} \mathbf{S}_t = \mathbf{U}\mathbf{S}_{t-1} + \mathbf{K}_t^\top \mathbf{V}_t \\ \boldsymbol{y}_t = \mathbf{Q}_t \mathbf{S}_t / \mathbf{Q}_t \mathbf{Z}_t + \mathbf{O}\boldsymbol{x}_t \end{cases} \tag{20}$$

where $\mathbf{U}$ denotes the identity matrix, $\mathbf{O}$ corresponds to the zero matrix, and $\boldsymbol{x}_t$ is the input token at step t. $\mathbf{S}_t$ represents the weighted sum of all past key–value pairs, and $\mathbf{Z}_t$ is the sum of all past keys. This iterative process accumulates the scanned information, and $\mathbf{Q}_t$ performs a "query" on the accumulated information to obtain the current output.

By comparing the formulations of Mamba and attention in (17) and (20), we observe the following correspondences: $\boldsymbol{h}_t$ and $\mathbf{S}_t$ represent one state; $\overline{\mathbf{A}}$ and $\mathbf{I}$ are the state transition matrices; $\mathbf{B}(\Delta \boldsymbol{x}_t)$ and $\mathbf{K}_t^\top \mathbf{V}_t$ denote the injection of new information. Notably, the output matrix $\mathbf{C}$ in Mamba plays a role analogous to the query ($\mathbf{Q}$) in attention, determining what information to extract from the memory. Based on this insight, we can modify $\mathbf{C}$ to incorporate information from unscanned sequences (i.e., global information), enabling Mamba to perform noncausal queries over the entire memory while retaining linear $O(L)$ complexity.

2) Attentive State-Space Equation: Based on the relationship between Mamba and attention mechanisms analyzed in Section III-B1, this section presents the detailed design of the ASE Module, which enables the construction of a noncausal Mamba model. The core idea is to inject global semantic information into the output matrix $\mathbf{C}$. To this end, a prompt mechanism is introduced to inject prompt vectors representing semantically similar pixel groups into $\mathbf{C} \in \mathbb{R}^{F\times d}$, enabling it to actively query pixels in the global image that are semantically similar to the current pixel. In this case, d denotes the hidden state dimension in the Mamba structure, while F is the length of the image sequence after flattening.

To store and query semantically similar information, we first construct a semantically decoupled prompt pool $\boldsymbol{\mathcal{P}} \in \mathbb{R}^{T\times d}$, which stores prompt vectors for groups of semantically similar pixels, with T being the number of prompt vectors in $\boldsymbol{\mathcal{P}}$. The semantic decoupling is formulated as

$$\boldsymbol{\mathcal{P}} = \mathbf{M}\mathbf{N} \tag{21}$$

where $\mathbf{N} \in \mathbb{R}^{r\times d}$ denotes the shared semantic feature space across all Mamba blocks, ensuring that all blocks generate prompts based on a unified "semantic language" to maintain global semantic consistency. $\mathbf{M} \in \mathbb{R}^{T\times r}$ is specific to each Mamba block, and the combination coefficients it contains act on the features in $\mathbf{N}$ to yield specific semantic prompt information. The internal rank $r \ll \min(T, d)$ compresses the high-dimensional hidden state space into a more compact semantic space, thereby reducing the number of parameters.

Subsequently, to match the prompts with the semantics of the currently processed pixels, a routing-based prompt selection mechanism is employed to assign the most relevant prompt vector to each position in the sequence. The specific operations are as follows: the flattened input features $\boldsymbol{x}_t \in \mathbb{R}^{F\times C}$ ($t = 1, 2, \ldots, F$) are first projected to the T-dimensional space via a linear layer, and the probability distribution of prompt selection is computed using LogSoftmax. To preserve the discrete selection property while enabling end-to-end training, the Gumbel-Softmax trick introduces a differentiable, approximate one-hot routing matrix $\mathbf{R} \in \mathbb{R}^{F\times T}$. Finally, the instance-specific prompt $\mathbf{P} = \mathbf{R}\boldsymbol{\mathcal{P}}$ is generated and integrated into matrix $\mathbf{C}$ via residual addition, yielding the ASE

$$\begin{cases} \boldsymbol{h}_t = \overline{\mathbf{A}}\boldsymbol{h}_{t-1} + \overline{\mathbf{B}}\boldsymbol{x}_t \\ \boldsymbol{y}_t = (\mathbf{C} + \mathbf{P})\,\boldsymbol{h}_t + \mathbf{D}\boldsymbol{x}_t. \end{cases} \tag{22}$$

This design enables the prompt pool to store representative semantic prototypes, and the routing mechanism dynamically matches dedicated prompts for each position. By injecting prompt information into matrix $\mathbf{C}$, its role is transformed into a "global semantic queryer", thereby effectively breaking the causal modeling constraint of Mamba and enabling noncausal global dependency modeling. Meanwhile, the method retains the linear computational complexity of state-space models and can be efficiently implemented via unidirectional scanning.

3) SGN Mechanism: Due to the causal modeling nature of Mamba, each step can only utilize information from tokens scanned previously, which inevitably affects the interaction between semantically similar but spatially distant pixels. Rather than simply unfolding images purely along the spatial dimension as in conventional approaches, we construct a semantic-level neighborhood via a new SGN mechanism. SGN explicitly brings semantically similar pixels closer in the unfolded sequence, enabling Mamba to focus its causal scanning process on informative and semantically relevant pixels.

This design builds upon the routing matrix $\mathbf{R}$ already learned in the ASE module, where $\mathbf{R}$ encodes the semantic prompt category assigned to each pixel. Leveraging this learned semantic information, we reorganize the image sequence accordingly. As illustrated in Fig. 1, the proposed *SGN-unfold* operation constructs a semantic-adjacent sequence by grouping pixels based on their prompt categories. Specifically, pixels assigned to the tth prompt category are aggregated to form the tth semantic group.

Subsequently, these semantic groups are concatenated (in ascending order of their category indices) to generate a semantic-adjacent sequence. In this reordered sequence, pixels with similar semantic attributes are positioned closer, effectively reducing the semantic distance along the sequence dimension.

Finally, the resulting semantic-adjacent sequence is fed into the proposed ASE module for state-space modeling, allowing Mamba to prioritize the aggregation of semantically correlated information during its causal scanning process. After state-space modeling, an *SGN-fold* operation (the inverse of SGN-unfold) is applied to reshape the sequence features back to the original 2-D spatial structure, producing the final output feature map.

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TABLE I
DETAILED SPECIFICATIONS OF THE FOUR DATASETS IN THIS STUDY

Dataset	Collection location	Collection equipment	Sample size	Image size	Classes	Spectral bands	Content	Train_ratio
PU	Pavia, Italy	Reflective Optics Spectrographic Imaging System sensor	42776	610×340	9	103	Urban	2%
CY	Jiangsu Province, China	a pushbroom hyperspectral imager	53734	348×512	10	80	Tea	2%
LK	Hubei Province, China	Head-wall Nanospectral imaging sensor	204542	550×400	9	270	Vegetation	1%
SV	Salinas Valley, California	Airborne Visible Infra-Red Imaging Spectrometer	53063	512×217	16	204	Vegetation	1%

IV. EXPERIMENTS

In this section, extensive experiments were conducted on four HSI datasets, and the performance of the proposed model was compared with the current state-of-the-art methods in the HSIC field. Ablation experiments were conducted to verify the effectiveness of each module in our model. Finally, a complexity analysis was performed to demonstrate the model's lightweight nature.

A. Datasets

To assess the effectiveness of the ARAM model, four widely used HSI datasets were employed: Pavia University (PU), Fanglu Tea Farm (CY), WHU-Hi-LongKou (LK), and Salinas Valley (SV). A detailed description of these datasets can be found in Table I.

B. Experimental Settings

1) Comparison Experiments: To comprehensively evaluate the performance of ARAM, it is benchmarked against the following state-of-the-art methods.

1) Machine learning-based
   a) *SVM [12]:* This model employs a support vector machine to achieve accurate classification of HSI.
2) CNN-Based
   a) *CNN3d [22]:* This method can simultaneously handle spatial and spectral features to enhance classification performance.
   b) *DSNet [49]:* This method adopts a dual-branch structure, designing a deep autoencoder to deeply fuse subpixel and pixel-level features.
3) Transformer-Based
   a) *SpectralFormer [27]:*— This network can concurrently capture both global and local spectral information from adjacent bands.
   b) *morphFormer [28]:* This model constructs a learnable morphological network by introducing morphological convolution operations in both spatial and spectral domains.
4) Mamba-Based
   a) *3DSS-Mamba [50]:* This model constructs a 3-D selective scanning mechanism, effectively extracting global features in high-dimensional space and improving computational efficiency.
   b) *S^2Mamba [34]*: This method combines spatial and spectral scanning mechanisms, achieving the extraction of spatial and spectral information.
   c) *HSI-MFormer [36]:* This model is a hybrid Mamba-Transformer framework that leverages the complementary strengths of both models.

2) Assessment Metrics: To measure the performance of the proposed ARAM model, this article adopts three widely used metrics in the field of HSIC: overall accuracy (OA), average accuracy (AA), and kappa coefficient.

3) Implementation Details: All experiments were conducted on an NVIDIA GeForce RTX 4070 GPU. To minimize randomness, the experimental results are reported as the average over 20 independent training runs, with each run conducted for 150 epochs. The comparative experiments were all conducted under the optimal parameters.

C. Parameter Analysis

To achieve optimal performance for the proposed ARAM model, extensive experiments were conducted to fine-tune several key parameters, including the number of PCA components, patch size, and hw_range.

1) Analysis of the Number of Principal Components in PCA: The high dimensionality of HSI data often results in considerable computational complexity. Therefore, PCA is commonly applied as a preprocessing technique to decrease dimensionality while preserving the most informative spectral features. The number of principal components in PCA specifies how many principal components are preserved during dimensionality reduction. A smaller number of components can result in the loss of crucial discriminative information, whereas an excessively large number limits the effectiveness of dimensionality reduction. Consequently, it is necessary to carefully adjust this parameter for different datasets. In our experiments, the number of retained principal components was set to {10, 15, 20, 25, 30}, and evaluations were conducted on four benchmark datasets. Fig. 3 displays the corresponding experimental results. For the PU, CY, and LK datasets, the best performance was reached with 15 principal components, while for the SV dataset, retaining 25 principal components yielded the highest classification accuracy. From the overall trend of each curve, the OA increases before reaching the optimal point due to the inclusion of more informative features, and then decreases as a result of the Hughes phenomenon [51].

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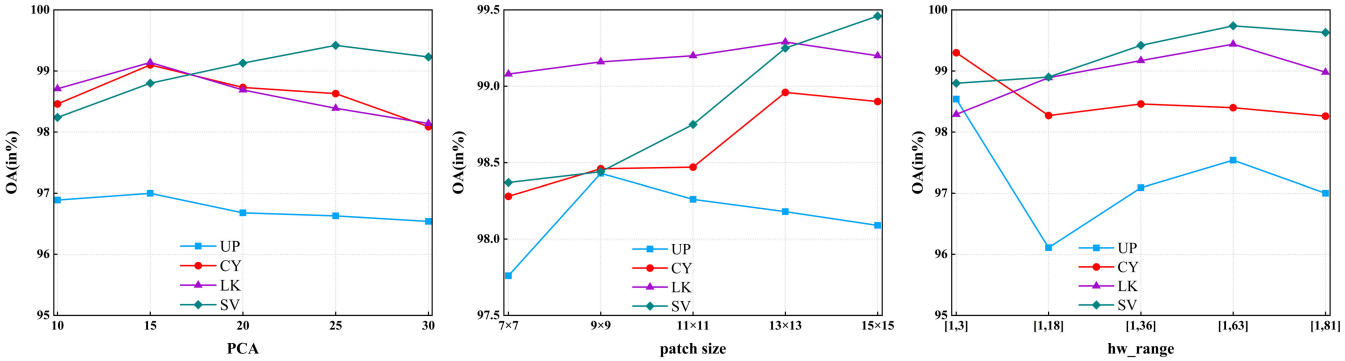


Fig. 3. Effects of different values of main parameters such as number of PCA components, patch size, and hw_range on the four datasets: PU, CY, LK, and SV.

2) Analysis of Different Patch Sizes: The patch size defines the spatial neighborhood window extracted around each central pixel for feature learning. A smaller patch size emphasizes local spatial details and incurs lower computational cost; however, it may fail to capture broader contextual relationships, leading to limited discriminative ability when dealing with complex landcover structures. In contrast, a larger patch size provides a richer spatial context for the model but results in a notable rise in computational cost. In our experiments, the patch size is varied from 7×7 to 15×15 with a step size of 2. As illustrated in Fig. 3, this parameter plays a crucial role in classification performance. Specifically, the optimal patch sizes are 9×9 for PU, 13×13 for CY and LK, and 15×15 for SV.

3) Analysis of Different Values of Hw_Range in ARConv: The parameter hw_range is a critical parameter in the proposed ARConv module. It defines the height and width range of the convolutional kernel, thereby determining both the spatial extent of the sampling window and the density of sampling points. Adjusting hw_range essentially reflects a trade-off between local detail preservation and global contextual modeling. In our experiments, hw_range is set to [1, 3], [1, 18], [1, 36], [1, 63], and [1, 81] to evaluate its impact on different datasets. As shown in Fig. 3, the SV and LK datasets achieve their best performance when hw_range is [1, 63], whereas the PU and CY datasets [1, 3]. This phenomenon directly reflects the inherent differences in spatial structures across datasets, indicating that different scenes impose distinct requirements on the spatial adaptability of convolutional kernels.

Specifically, the SV and LK datasets contain large homogeneous regions and exhibit pronounced scale variations, which demand stronger long-range dependency modeling. A larger hw_range enables ARConv to adaptively generate larger and more flexible sampling windows, thereby enhancing its capability to extract long-range spatial information. By comparison, the PU and CY datasets are characterized by relatively compact object distributions and smaller target scales, where dense and localized sampling with a smaller hw_range is more effective for capturing fine-grained local spatial features. These results demonstrate that the ARConv module can adaptively respond to varying spatial complexities across datasets, highlighting its flexibility and effectiveness in diverse HSIC scenarios.

D. Comparison With State-of-the-Art Methods

To provide a fair evaluation of ARAM, we perform comparative experiments with multiple state-of-the-art HSIC networks on the four considered benchmark datasets: PU, CY, LK, and SV. To ensure a fair comparison, all competing methods are carefully fine-tuned to their optimal parameter settings. From the experimental results, the performance of different methods is analyzed in both quantitative and qualitative terms. Our comprehensive evaluation demonstrates that ARAM achieves the best performance in terms of both quantitative metrics and visual results.

1) Quantitative Analysis: Tables II–V report the quantitative comparison results of all methods on the four datasets. It is clearly observable that the proposed ARAM attains superior OA, AA, and Kappa on all datasets, demonstrating its strong capability in land cover classification. Machine learning methods, due to their lack of global modeling capabilities, fail to fully exploit the rich spatial–spectral information contained in HSIs, leading to suboptimal performance. The CNN3d model, constrained by its fixed receptive field size, lacks adaptive capacity when faced with complex HSI data, leaving considerable room for improvement in hyperspectral image classification tasks. In contrast, DSNet captures both superpixel and pixel-level features, outperforming CNN3d. However, its lack of global modeling capability results in lower performance compared to ours. Regarding spectral–spatial features, compared to SpectralFormer, morphFormer improves the extraction and attention to spatial features in HSIs, achieving better classification results on the four datasets, with OA improvements of 5.70%, 2.09%, 0.44%, and 5.84%, respectively. As a result of the powerful long-sequence modeling capability of SSM, Mamba-based variants such as 3DSS-Mamba and S^2Mamba, also exhibit strong competitiveness. Yet, due to the causal sequencing of Mamba, its performance is still limited. Meanwhile, HSI-MFormer achieves a significant performance boost by combining the advantages of Mamba and Transformer. Despite this, the proposed ARAM still outperforms this model in overall performance. This is mainly

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TABLE II

OA, AA, AND KAPPA OBTAINED BY DIFFERENT APPROACHES ON THE PU DATASET. THE HIGHEST RESULT IN EACH ROW IS BOLDED

Class	SVM	CNN3d	DSNet	SpectralFormer	morphFormer	3DSS-Mamba	S^{2}Mamba	HSI-MFormer	Proposed
Asphalt	92.50	94.73	98.24	91.57	98.84	96.72	**99.26**	97.20	98.07
Meadows	96.25	98.70	99.76	98.31	99.88	98.99	99.82	99.76	**100.00**
Gravel	70.30	62.94	91.60	77.95	91.86	93.09	92.05	93.93	**94.98**
Trees	84.67	96.36	96.13	93.96	**97.79**	92.21	96.18	91.30	96.56
Mental sheets	98.98	99.54	99.97	**100.00**	**100.00**	99.47	99.86	99.08	99.46
Bare soil	74.13	79.25	98.81	86.27	98.73	95.78	97.94	99.77	**99.82**
Bitumen	54.60	70.86	93.55	60.91	95.53	81.57	87.78	**100.00**	97.89
Bricks	84.20	93.19	95.09	88.30	93.89	93.33	**98.74**	98.34	93.51
Shadow	99.81	99.05	99.32	95.59	91.20	97.64	94.02	95.71	**99.38**
OA(%)	88.80±0.83	92.57±0.81	98.16±0.23	92.50±0.43	98.20±0.41	96.44±2.04	98.27±0.17	98.35±0.20	**98.54±0.18**
AA(%)	83.94±1.60	88.29±1.34	96.86±0.52	88.09±1.04	96.41±1.29	94.31±2.66	96.18±0.34	97.23±0.34	**97.80±0.30**
Kappa(%)	84.97±1.08	90.05±1.08	97.56±0.30	90.01±0.59	97.61±0.55	95.26±2.76	97.71±0.22	97.81±0.26	**98.07±0.24**

TABLE III

OA, AA, AND KAPPA OBTAINED BY DIFFERENT APPROACHES ON THE CY DATASET. THE HIGHEST RESULT IN EACH ROW IS BOLDED

Class	SVM	CNN3d	DSNet	SpectralFormer	morphFormer	3DSS-Mamba	S^{2}Mamba	HSI-MFormer	Proposed
Masson pine	92.76	96.58	99.11	98.32	99.51	97.67	99.20	99.69	**99.74**
Bamboo forest	27.80	57.11	90.43	84.74	91.49	86.33	91.79	98.98	**99.04**
Tea tree	97.97	98.21	99.53	98.69	99.44	98.91	99.14	99.74	**99.78**
Reed	10.86	63.27	93.71	98.44	94.90	77.48	78.24	**99.34**	93.38
Rice	98.41	99.46	99.76	98.43	99.87	99.64	**99.99**	99.97	99.91
Sweet potato	87.29	83.09	90.77	81.63	**97.70**	91.67	90.51	92.87	85.89
Coriander	67.51	87.67	93.40	88.16	97.07	85.10	77.77	**100.00**	96.93
Waste grass	45.04	59.16	88.70	66.65	89.12	79.47	82.24	92.70	**92.83**
Ponds	99.53	99.71	99.80	99.90	99.76	99.44	99.90	98.80	**99.92**
Building/road	93.41	98.43	98.01	97.46	99.23	91.91	**99.38**	77.70	93.62
OA(%)	91.15±0.68	94.79±0.19	98.54±0.18	96.68±0.33	98.77±0.11	97.29±0.78	98.07±0.23	98.99±0.14	**99.31±0.13**
AA(%)	72.09±5.70	84.27±1.53	95.32±0.98	91.24±1.89	96.81±0.45	90.76±4.10	91.81±0.99	95.71±0.73	**97.55±0.25**
KAPPA(%)	87.87±1.11	92.22±0.27	97.84±0.27	95.07±0.50	98.18±0.16	95.98±1.17	97.14±0.34	98.50±0.20	**98.97±0.19**

TABLE IV

OA, AA, AND KAPPA OBTAINED BY DIFFERENT APPROACHES ON THE LK DATASET. THE HIGHEST RESULT IN EACH ROW IS BOLDED

Class	SVM	CNN3d	DSNet	SpectralFormer	morphFormer	3DSS-Mamba	S^{2}Mamba	HSI-MFormer	Proposed
Corn	99.12	99.55	99.88	99.74	99.55	99.69	99.89	**99.92**	99.80
Cotton	89.37	96.08	98.48	94.60	98.47	99.37	97.65	99.26	**99.48**
Sesame	83.07	86.32	91.80	96.15	96.47	95.24	94.08	96.46	**97.26**
Broad-leaf soybean	98.02	98.80	99.43	98.67	99.57	99.40	99.61	**99.79**	99.67
Narrow-leaf soybean	71.57	83.00	93.74	87.02	92.15	96.96	92.76	96.47	**97.82**
Rice	99.46	99.54	99.73	99.53	99.61	98.81	99.62	**99.77**	99.30
Water	99.93	99.98	**99.99**	99.98	99.98	99.87	99.92	99.95	99.89
Roads and houses	88.27	**97.53**	96.79	95.95	96.83	96.14	96.10	95.51	96.16
Mixed weed	82.11	94.07	95.42	93.53	89.47	89.81	94.49	92.52	**95.72**
OA(%)	97.06±0.36	98.57±0.21	99.25±0.10	98.66±0.11	99.11±0.18	99.07±0.15	99.20±0.11	99.34±0.03	**99.44±0.08**
AA(%)	96.12±0.48	94.99±0.92	97.25±0.82	96.13±0.69	96.90±0.91	97.25±0.63	97.12±0.61	97.74±0.13	**98.34±0.33**
KAPPA(%)	90.10±1.72	98.12±0.27	99.01±0.13	98.24±0.14	98.83±0.24	98.82±0.20	98.95±0.14	99.20±0.04	**99.26±0.10**

because our model not only extracts more detailed features but also mitigates the causality of Mamba through AM, enhancing the model's global modeling capabilities.

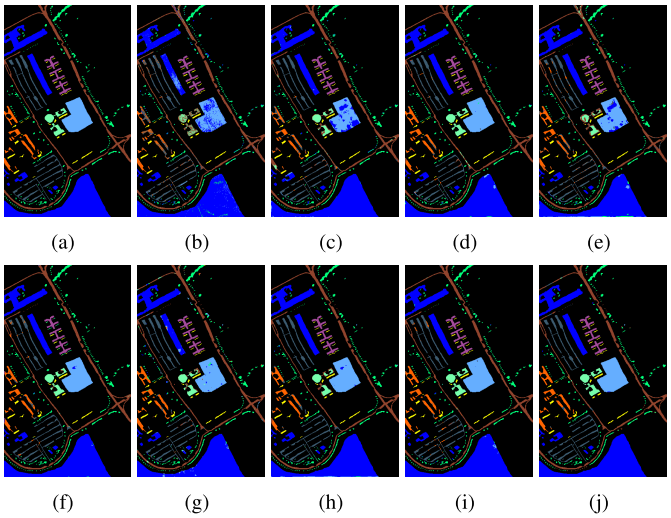
Further analysis from the perspective of per-class classification accuracy reveals that the ARAM model achieves the highest classification accuracy most frequently across different classes. For example, on the CY dataset, ARAM outperforms the other methods in five classes, while the second-best models, S^{2}Mamba and HSI-MFormer, perform best in only 2 classes, and MorphFormer achieves the best performance in 1 class. For the SV dataset, ARAM delivers the top classification accuracy across 8 classes, far surpassing the second-place HSI-MFormer. These results indicate that ARAM can effectively alleviate interclass confusion and enhance discrimination for hard-to-classify land-cover categories. Overall, by effectively integrating local and global information, the proposed ARAM consistently demonstrates superior classification performance

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TABLE V

OA, AA, AND KAPPA OBTAINED BY DIFFERENT APPROACHES ON THE SV DATASET. THE HIGHEST RESULT IN EACH ROW IS BOLDED

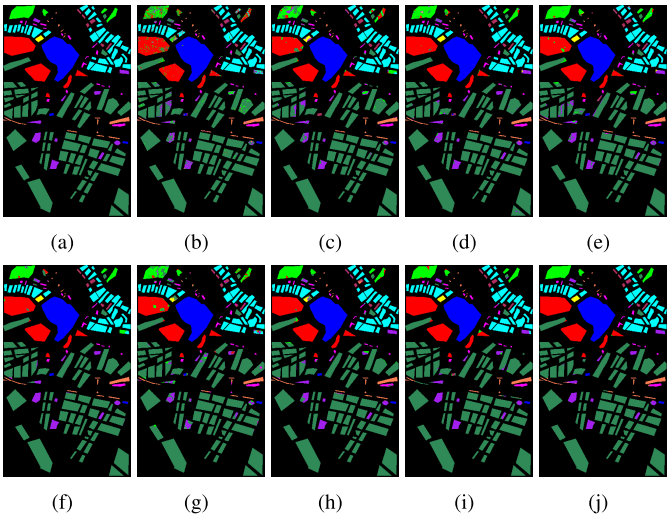
Class	SVM	CNN3d	DSNet	SpectralFormer	morphFormer	3DSS-Mamba	S^2Mamba	HSI-MFormer	Proposed
Brocoli-green-weeds-1	97.08	97.90	99.96	86.58	98.17	98.00	93.35	**100.00**	99.85
Brocoli-green-weeds-2	99.38	99.86	99.33	99.49	99.90	99.34	99.57	**100.00**	**100.00**
Fallow	96.27	85.76	96.80	93.01	96.35	98.30	91.95	99.80	**99.89**
Fallow-rough-plow	99.15	98.70	98.55	94.38	96.33	97.08	98.83	**99.46**	97.53
Fallow-smooth	96.89	93.09	95.45	86.97	96.27	96.69	97.98	**99.85**	98.73
Stubble	99.24	99.99	99.93	99.95	**100.00**	99.40	**100.00**	99.90	99.65
Celery	99.20	99.69	99.89	96.29	99.48	96.83	99.82	99.71	**99.88**
Grapes-untrained	89.28	84.67	90.87	85.38	93.54	94.30	90.64	99.16	**99.96**
Soil-vinyard-develop	99.26	98.72	99.69	98.72	99.30	99.70	99.95	**100.00**	**100.00**
Corn-senesced-green-weeds	90.03	89.87	96.05	86.14	96.63	93.39	96.87	99.03	**99.91**
Lettuce-romaine-4wk	90.04	69.82	92.38	79.58	94.15	94.03	65.44	**100.00**	99.41
Lettuce-romaine-5wk	99.34	98.68	99.45	98.15	99.61	92.57	**99.96**	99.57	99.40
Lettuce-romaine-6wk	98.77	99.96	**99.97**	96.41	99.33	89.89	99.13	97.10	88.88
Lettuce-romaine-7wk	88.60	99.08	98.98	95.96	96.68	92.15	**99.33**	99.19	97.54
Vinyard-untrained	56.77	73.54	89.84	77.46	88.26	88.91	85.55	96.88	**99.09**
Vinyard-vertical-trellis	95.17	88.94	96.79	90.36	98.87	91.87	97.57	98.60	**99.98**
OA(%)	90.08±0.28	90.46±0.85	95.72±0.43	90.19±0.88	96.03±0.46	95.24±1.01	94.41±0.69	99.23±0.14	**99.41±0.25**
AA(%)	88.92±0.31	92.39±1.23	97.12±0.73	91.55±0.72	97.05±0.67	95.15±1.60	94.75±1.80	98.81±0.10	**98.92±0.77**
KAPPA(%)	93.40±0.41	89.37±0.95	95.24±0.48	89.07±0.98	95.58±0.51	94.69±1.13	93.77±0.77	99.14±0.16	**99.35±0.28**



(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Fig. 4. Classification maps obtained by different methods on the PU dataset. (a) Ground truth, (b) SVM, OA = 88.80%, (c) CNN3d, OA = 92.57%, (d) DSNet, OA = 98.16%, (e) SpectralFormer, OA = 92.50%, (f) morphFormer, OA = 98.20%, (g) 3DSS-Mamba, OA = 96.44%, (h) S^2Mamba, OA = 98.27%, (i) HSI-MFormer, OA = 98.35%, and (j) proposed, OA = 98.54%.



(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Fig. 5. Classification maps obtained by different methods on the CY dataset. (a) Ground truth, (b) SVM, OA = 91.15%, (c) CNN3d, OA = 94.79%, (d) DSNet, OA = 98.54%, (e) SpectralFormer, OA = 96.68%, (f) morphFormer, OA = 98.77%, (g) 3DSS-Mamba, OA = 97.29%, (h) S^2Mamba, OA = 98.07%, (i) HSI-MFormer, OA = 98.99%, and (j) proposed, OA = 99.31%.

across different datasets and land-cover categories, thereby validating its effectiveness and advancements in HSIC tasks.

2) Qualitative Analysis: To present a more intuitive comparison of the classification performance of different methods, the classification maps obtained on the four datasets are visualized in Figs. 4–7. By examining the presence of classification noise, artifacts, small isolated regions, and boundary delineation, it is evident that the proposed ARAM exhibits enhanced classification accuracy compared with the competing methods. Fig. 4 illustrates the classification results on the PU dataset. In comparison with other methods, HSI-MFormer and the proposed ARAM perform the best. However, upon closer inspection, the result from HSI-MFormer shows a small amount of orange noise in the lower-left corner. In Fig. 5, which corresponds to the CY dataset, the object distribution is relatively scattered. However, the green region in the upper-left corner clearly highlights the advantage of ARAM. For the LK dataset, Fig. 6 reveals that the classification maps from SVM, CNN3d, DSNet, SpectralFormer, and morphFormer exhibit significant noise and small patches. Moreover, the remaining models, 3DSS-Mamba, S^2Mamba, and HSI-MFormer, show noticeable misclassifications near the boundary of the yellow region on the left. In contrast, ARAM achieves better object region integrity and clearer boundaries. A more striking obser-

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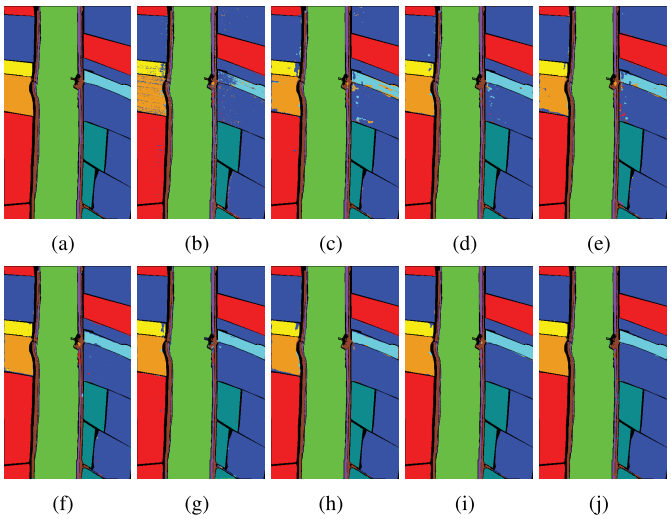


Fig. 6. Classification maps obtained by different methods on the LK dataset. (a) Ground truth, (b) SVM, OA = 97.06%, (c) CNN3d, OA = 98.57%, (d) DSNet, OA = 99.25%, (e) SpectralFormer, OA = 98.66%, (f) morphFormer, OA = 99.11%, (g) 3DSS-Mamba, OA = 99.07%, (h) S^{2}Mamba, OA = 99.20%, (i) HSI-MFormer, OA = 99.34%, and (j) proposed, OA = 99.44%.

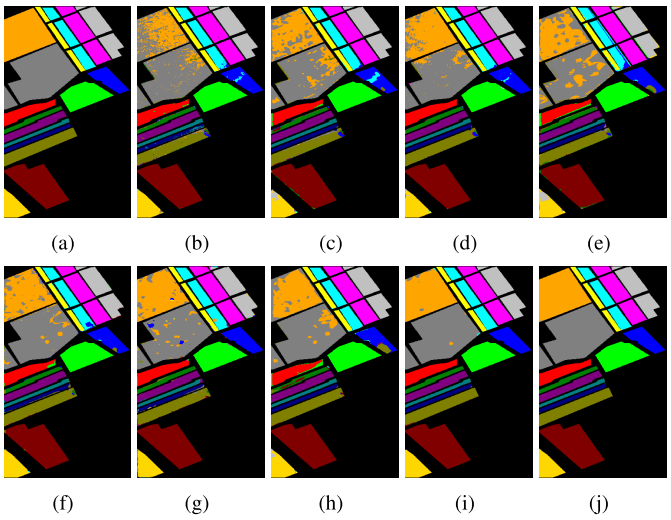


Fig. 7. Classification maps obtained by different methods on the SV dataset. (a) Ground truth, (b) SVM, OA = 90.08%, (c) CNN3d, OA = 90.46%, (d) DSNet, OA = 95.72%, (e) SpectralFormer, OA = 90.19%, (f) morphFormer, OA = 96.03%, (g) 3DSS-Mamba, OA = 95.24%, (h) S^{2}Mamba, OA = 94.41%, (i) HSI-MFormer, OA = 99.23%, and (j) proposed, OA = 99.41%.

vation is in the SV dataset, where, except for our method, all the comparison methods in this study show confusion in the classification of the "Vinyard-untrained" and "Grapes-untrained" classes. In conclusion, the qualitative visualization results further validate the superiority of the proposed model.

E. Ablation Study

1) Ablation Study on ARAM: To evaluate the contribution of the key components of the proposed model, including the ARConv module and the two AM submodules (ASE and SGN), ablation experiments were carried out on four datasets: PU, CY, LK, and SV. For a fair comparison, standard convolution was used in place of ARConv upon removal of the ARConv module; the basic Mamba architecture was applied following the exclusion of the ASE module; and the semantic-based image reorganization was disabled when the SGN module was removed.

TABLE VI

ABLATION STUDY ON THE CORE COMPONENTS OF OUR APPROACH: ARCONV, ASE, AND SGN

MODULE			DATASET			
ARConv	ASE	SGN	PU	CY	LK	SV
✓			97.37	98.47	97.78	99.14
	✓		97.20	98.35	97.65	99.02
		✓	97.15	98.28	97.59	98.95
✓	✓		97.54	98.91	98.85	99.26
	✓	✓	98.30	98.63	99.33	99.15
✓		✓	97.42	98.80	98.79	99.21
✓	✓	✓	**98.54**	**99.31**	**99.44**	**99.41**

TABLE VII

ABLATION STUDY ON THE ARCONV MODULE

MODULE	DATASET			
	PU	CY	LK	SV
Deformable_conv	97.73	97.03	97.23	96.98
Multi-scale_conv	95.11	96.78	97.73	97.24
Standalone ARConv	98.09	99.00	98.36	99.27
Residual ARConv	**98.54**	**99.31**	**99.44**	**99.41**

Table VI reports the ablation results, demonstrating the positive effect of each module on classification accuracy. Specifically, the introduction of the ARConv module improves the OA by 0.24%, 0.68%, 0.26%, and 0.11% on the four datasets, respectively, owing to ARConv's excellent ability to extract fine-grained local features. The ASE module further boosts the OA by 1.12%, 0.51%, 0.20%, and 0.65%, while the SGN module yields additional OA gains of 1.00%, 0.40%, 0.15%, and 0.59%. This is because the proposed ASE and SGN modules can query and reorganize information from semantically similar pixels, mitigating the long-range attenuation problem and enhancing effective interactions between distant similar features, thereby improving recognition results. These results clearly demonstrate that the ARConv, ASE, and SGN modules exhibit strong complementarity and possess powerful feature extraction capabilities.

2) Ablation Study on ARConv Module: To verify the necessity and superiority of the residual ARConv module, ablation experiments are conducted on four hyperspectral datasets, including PU, CY, LK, and SV. Comparative experiments are designed for four convolution structures: Deformable_conv, Multiscale_conv, standalone ARConv, and residual ARConv. It is worth noting that the ASE and SGN modules in ARAM are fixed and retained throughout the experiment, and only the convolution module is replaced to ensure a single variable in the experimental setup.

The experimental results in Table VII indicate that traditional Multiscale_conv and Deformable_conv achieve relatively low classification accuracy, and exhibit limited capability in modeling multiscale differences and geometric

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TABLE VIII
COMPLEXITY ANALYSIS OF DIFFERENT COMPARATIVE METHODS ON THE PU DATASET (PARAMETERS, FLOPS, TRAINING TIME, AND TESTING TIME)

Metrics	SVM	CNN3d	DSNet	SpectralFormer	morphFormer	3DSS-Mamba	S^2Mamba	HSI-MFormer	Proposed
Params(K)	–	246.7190	651.8490	178.2170	148.9050	24.1400	76.7330	170.4000	42.7000
Flops(G)	–	0.0679	0.0131	0.0159	0.0216	10.1500	0.0093	0.0742	0.0020
Training time(s)	0.2587	38.1010	33.1110	106.5040	106.5410	61.1980	129.5080	35.3900	73.4000
Testing time(s)	1.2601	6.8120	8.1160	20.7100	35.7170	6.1130	11.8080	5.6700	11.7300

deformation features of hyperspectral images. Compared with the two conventional convolutions, the standalone ARConv without residual connections obtains a substantial accuracy improvement across all four datasets, which demonstrates that the basic ARConv structure proposed in this article possesses superior capability in fine-grained feature extraction and adaptive feature modeling. On this basis, the residual ARConv with residual connections further outperforms all comparative convolution schemes and achieves the optimal accuracy on the PU, CY, LK, and SV datasets. This fully validates that the introduction of residual structure can optimize feature propagation and alleviate the degradation problem of deep networks. Meanwhile, it enables the ARConv module to fully exert its advantages in multiscale adaptation and geometric deformation modeling, thereby further verifying the necessity and superiority of the proposed residual ARConv module.

F. Complexity Analysis

In this section, we further perform a statistical analysis and comparison of the proposed ARAM with all baseline methods on the PU dataset in terms of the number of parameters, floating-point operations (FLOPs), as well as training time and testing time. Table VIII lists the corresponding results. In terms of model size, ARAM contains only 42.7K parameters, which is substantially smaller than most competing methods, including CNN3d (246.7K), DSNet (651.8K), SpectralFormer (178.2K), MorphFormer (148.9K), and HSI-MFormer (170.4K). Compared with the lightweight model 3DSS-Mamba (24.1K), ARAM introduces only a modest increase in parameters, while achieving significantly better classification accuracy, thereby striking a more favorable balance between model capacity and performance. Moreover, the computational cost of ARAM is merely 0.0022G FLOPs, which is the lowest among all compared methods, demonstrating its high computational efficiency and lightweight nature. In addition, ARAM also exhibits superior performance in both training and inference time. Overall, the proposed ARAM achieves an optimal balance between computational complexity and classification performance.

V. CONCLUSION

This article proposes a new ARAM model for HSIC. Our model enhances the extraction of local fine-grained features while mitigating the long-range attenuation problem, capturing more global contextual information, all while maintaining low computational complexity. By means of the newly designed ARConv module, our model can adaptively extract richer local details, compensating for Mamba's limitations in capturing fine-grained local features. Regarding the attention mechanisms, the developed ASE module effectively mitigates long-range attenuation while the new SGN mechanism enhances interactions among related pixels, working synergistically to achieve semantic-aware global modeling. Extensive evaluations across four public benchmark datasets reveal that ARAM surpasses existing state-of-the-art methods in terms of performance. Moreover, the proposed model attains excellent classification results with minimal FLOPs (0.0022G) and a compact parameter size, highlighting its efficiency and practicality. Future research will focus on exploring optimized integrations with different network architectures and expanding the proposed framework to other HSI processing tasks.

REFERENCES

[1] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.
[2] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 8723–8737, Sep. 2025.
[3] L. Liang et al., "S2PW-mamba: Pinwheel and wavelet-based spatial–spectral mamba for hyperspectral image classification," *Expert Syst. Appl.*, vol. 324, Aug. 2026, Art. no. 132497.
[4] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.
[5] H. Yang, H. Yu, K. Zheng, J. Hu, T. Tao, and Q. Zhang, "Hyperspectral image classification based on interactive transformer and CNN with multilevel feature fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[6] L. Liang et al., "LKMA: Learnable kernel and mamba with spatial–spectral attention fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5530914.
[7] J. Jia, H. Yu, C. Wang, K. Zheng, J. Li, and J. Hu, "Spectral–spatial collaborative pretraining framework with multiconstraint cooperation for hyperspectral–multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11610–11622, 2025.
[8] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Subpixel spectral variability network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504014.
[9] M. Shimoni, R. Haelterman, and C. Perneel, "Hyperspectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.
[10] B. Zhang, Y. Chen, Z. Li, S. Xiong, and X. Lu, "SANet: A self-attention network for agricultural hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501315.
[11] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.
[12] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

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[13] C. Rodarmel and J. Shan, "Principal component analysis for hyperspectral image classification," *Surveying Land Inf. Sci.*, vol. 62, no. 2, pp. 115–122, 2002.

[14] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[15] J. Xia, P. Ghamisi, N. Yokoya, and A. Iwasaki, "Random forest ensembles and extended multiextinction profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 202–216, Jan. 2018.

[16] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[17] Y. Gu, T. Liu, X. Jia, J. A. Benediktsson, and J. Chanussot, "Nonlinear multiple kernel learning with multiple-structure-element extended morphological profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3235–3247, Jun. 2016.

[18] Y. Duan, H. Huang, and T. Wang, "Semisupervised feature extraction of hyperspectral image using nonlinear geodesic sparse hypergraphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515115.

[19] Y. Zhang, W. Li, W. Jia, M. Zhang, R. Tao, and S. Liang, "Cross-domain hyperspectral image classification based on bi-directional domain adaptation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 12038–12051, Dec. 2025.

[20] W. An, Z. Feng, and S. Yang, "Progressive multiscale generator for domain generalization in hyperspectral image classification with small sample," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 6, pp. 7790–7803, Jun. 2026.

[21] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3322–3325.

[22] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[23] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "Feedback attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501916.

[24] M. Zhu, J. Fan, Q. Yang, and T. Chen, "SC-EADNet: A self-supervised contrastive efficient asymmetric dilated network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519517.

[25] I. Ahmad et al., "A multiscale transformer with spatial attention for hyperspectral image classification," *Sci. Rep.*, vol. 16, no. 1, Jan. 2026.

[26] B. Zhang, Y. Chen, S. Xiong, and X. Lu, "Hyperspectral image classification via cascaded spatial cross-attention network," *IEEE Trans. Image Process.*, vol. 34, pp. 899–913, 2025.

[27] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[28] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[29] Y. He, B. Tu, B. Liu, Y. Chen, J. Li, and A. Plaza, "Hybrid multi-scale spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527918.

[30] L. Liang et al., "Cross-stage attention edge enhancement and Fourier-wavelet transformer integrated network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 28797–28811, 2025.

[31] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Model.*, 2024.

[32] Z. Han et al., "Multisource collaborative domain generalization for cross-scene remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535815.

[33] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[34] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S^2mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511413.

[35] L. Liang et al., "DBMLLA: Double-branch mamba-like linear attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524315.

[36] A. Du et al., "Cross-domain hyperspectral image classification via mamba–CNN and knowledge distillation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524415.

[37] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "HSI-MFormer: Integrating mamba and transformer experts for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621916.

[38] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[39] R. Zhang, S. Tang, Y. Zhang, J. Li, and S. Yan, "Scale-adaptive convolutions for scene parsing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2050–2058.

[40] Y. Qi, Y. He, X. Qi, Y. Zhang, and G. Yang, "Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6070–6079.

[41] H. Su, V. Jampani, D. Sun, O. Gallo, E. Learned-Miller, and J. Kautz, "Pixel-adaptive convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 11166–11175.

[42] M. S. A. M. Al-Gaashani, R. Alkanhel, M. A. S. Ali, M. S. A. Muthanna, A. Aziz, and A. Muthanna, "MSCPNet: A multi-scale convolutional pooling network for maize disease classification," *IEEE Access*, vol. 13, pp. 11423–11446, 2025.

[43] I. Cosmin Duta, L. Liu, F. Zhu, and L. Shao, "Pyramidal convolution: Rethinking convolutional neural networks for visual recognition," 2020, *arXiv:2006.11538*.

[44] Y. Yin, Z. Han, M. Jian, G.-G. Wang, L. Chen, and R. Wang, "AMSUnet: A neural network using atrous multi-scale convolution for medical image segmentation," *Comput. Biol. Med.*, vol. 162, Aug. 2023, Art. no. 107120.

[45] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[46] G. Pechlivanidou and N. Karampetakis, "Zero-order hold discretization of general state space systems with input delay," *IMA J. Math. Control Inf.*, vol. 39, no. 2, pp. 708–730, Jun. 2022.

[47] E. J. Kirkland, "Bilinear interpolation," in *Advanced Computing in Electron Microscopy*. Cham, Switzerland: Springer, 2010, pp. 261–263.

[48] C. Ge et al., "Demystify mamba in vision: A linear attention perspective," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 127181–127203.

[49] S. Feng et al., "DSNet: Dynamic stitchable neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5519214.

[50] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[51] W. Ma, C. Gong, Y. Hu, P. Meng, and F. Xu, "The Hughes phenomenon in hyperspectral classification based on the ground spectrum of grasslands in the region around Qinghai lake," *Proc. SPIE*, vol. 8910, pp. 363–373, Aug. 2013.

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